

ECO387C - Fall 2024

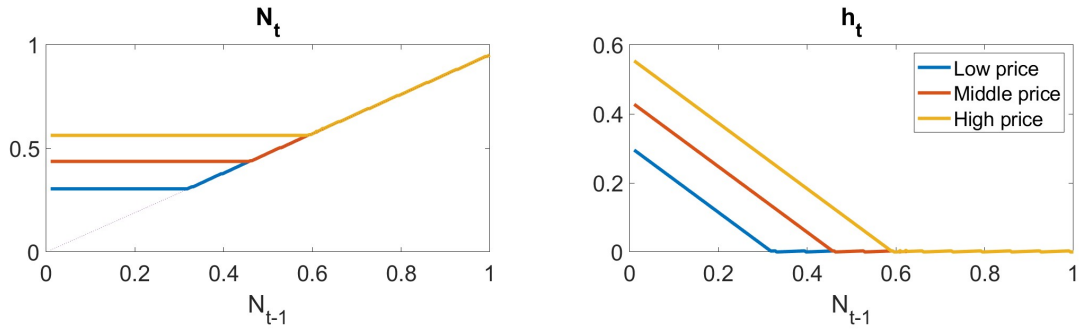
Problem Set 5 Solutions

Raymond He & Husang Kim

1. The relevant state variables are N_{t-1} and p_t (N and p).
2. The Bellman equation is

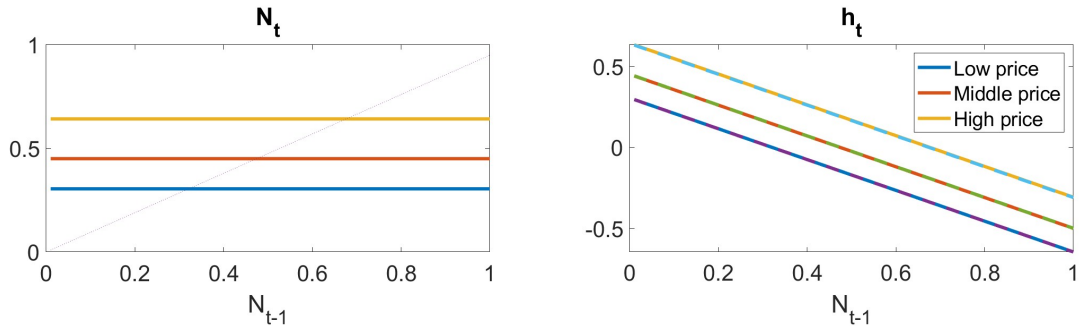
$$V(N, p) = \max_{N' \geq N} \left\{ p(N')^{(1-\alpha)} - w(N') + \frac{1}{1+r} \mathbb{E}_{p'|p} [V(N', p')] \right\}$$

3. The policy function with constraint is in the figure below:



Given some N_{t-1} , when the price is higher, the firms tend to hire more (larger h_t). After some threshold, however, the firms choose not to hire anymore ($h_t = 0$) and let workers leave each period.

4. The policy function without constraint is in the figure below:



Given some N_{t-1} , when the price is higher, the firms still tend to hire more and retain a larger workforce (larger h_t and N_t). The left panel demonstrates that for each price level,

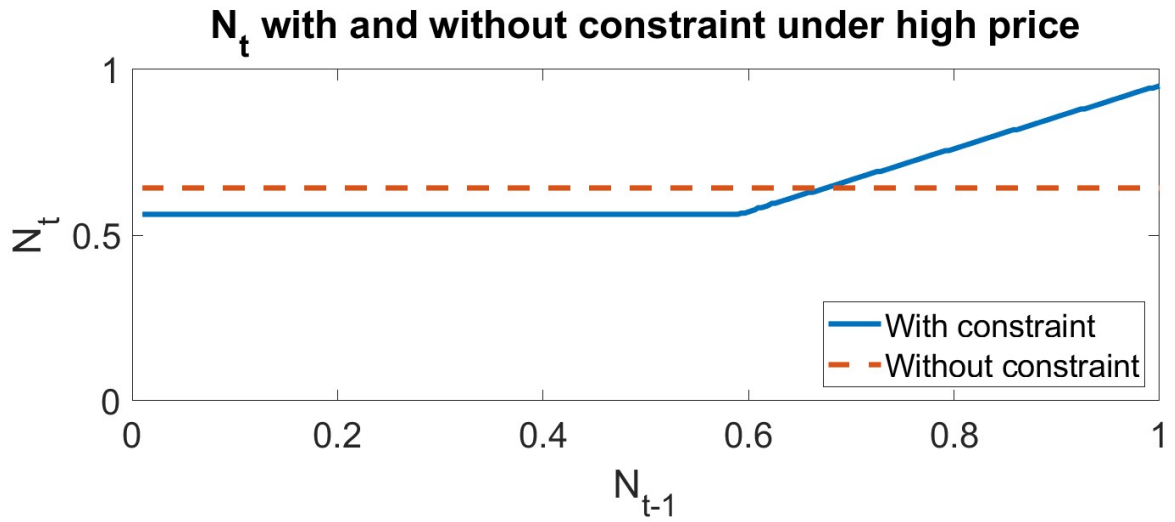
there is an optimal workforce level (N_t). Without the constraint, the firm can adjust h_t freely in each period to achieve that optimal workforce level.

Note also that taking the first order condition with respect to h_t and setting it equal to zero gives the policy function:

$$h_t = \left(\frac{(1-\alpha)p_t}{w} \right)^{1/\alpha} - (1-\delta)N_{t-1}$$

This policy function for each price level is overlayed with the numerical policy functions obtained from the value function iteration in the right panel for each price level.

The other effect of constraint can be illustrated by comparing the numerical policy functions with and without the hiring constraint. For example, the figure below compares the policy function N_t with and without constraint under the highest price (p_3).



The red dashed line indicates the optimal level of workforce the firm would achieve without the constraint. The blue line indicates the level of workforce the firm tries to achieve when the constraint is introduced. At the low level of a given workforce (e.g. $N_{t-1} = 0.2$) The firm under constraint chooses the workforce level N_t that falls short of the optimal level without constraint. This is because the firm internalizes that it cannot freely fire workers and hence becomes reluctant to achieve the optimal level of the workforce. The area where the blue line is upward-sloping indicates the region where the constrained firm would choose $h_t = 0$ and let workers leave.